

Induction Behind Regularisation / 2 (weight)

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regularization

Only to weights, can't be done on bias

$$W_n = W_0 - \eta \left(\frac{\partial L}{\partial W_0} + \lambda W_0 \right) \quad \text{--- (I)}$$

Learning rate (hyperparameter)

without regularization

After Regularisation

Change in Loss function

$$L' = L + \frac{\lambda}{2} \sum_i |W_i|^2$$

λ (hyperparameter) 1/2 regularization

not for loss function (sample) $(W_0^2 + W_1^2 + \dots)$

So,

$$\frac{\partial L'}{\partial W_0} = \frac{\partial L}{\partial W_0} + \frac{1}{2} \cdot 2 W_0$$

$$\text{or, } \frac{\partial L'}{\partial W_0} = \frac{\partial L}{\partial W_0} + W_0$$

New weight,

$$W_n = W_0 - \eta \left(\frac{\partial L'}{\partial W_0} + W_0 \right)$$

use L2 reg
use L2 → deep learning overfits issue

$$\text{or, } W_n = W_0 - \eta \left(\frac{\partial L}{\partial W_0} + \lambda W_0 \right)$$

$$\text{or } W_n = W_0 - \eta \lambda W_0 - \eta \frac{\partial L}{\partial W_0}$$

$$\text{or, } W_n = ((1 - \eta \lambda) W_0) - \eta \frac{\partial L}{\partial W_0} \quad \text{--- (II)}$$

Comparing (I) & (II)

λ ≥ 0 ← positive

$(1 - \eta \lambda) W_0$ & W_0

New weight after regularization

without regularization

Weight decay factor (the shrink multiplier)

$$(1 - \eta \lambda) W_0$$

positive

So, new $W_0 < W_0$ using regularization (without regularization)

Thus, W keeps decreasing & W will close to zero but never reach zero